

b) We shall prove $(p \rightarrow q)$ is True by contradiction.

$\therefore$ We shall prove $(p \wedge \neg q) \rightarrow F$ is True.

Let $(p \wedge \neg q)$ be True. By defn. of AND, p is True and $\neg q$ is True.

From part (a), $3n+2$ is an even integer and n is an odd integer

By defn 1, there exists integers k_1, k_2 s.t. $(3n+2) = 2k_1$ and $n = 2k_2 + 1$

$$(3n+2) - n = 2k_1 - (2k_2 + 1) \Rightarrow (2n+2) = 2(k_1 - k_2 - 1) + 1$$

$$\Rightarrow 2(n - k_1 + k_2 + 1) = 1 \Rightarrow \boxed{2(n - k_1 + k_2 + 1) = 1} \Rightarrow \boxed{(n - k_1 + k_2 + 1) = \frac{1}{2}}$$

We know, $\because n, k_1$ and k_2 are integers, $n - k_1 + k_2 + 1$ is an integer.

But $\frac{1}{2}$ is not an integer. This is a contradiction.

$\therefore \neg q$ is False. By defn. of AND, $(p \wedge \neg q)$ is False. By defn. of ~~AND~~ conditional,

$(p \wedge \neg q) \rightarrow F$ is True. $\therefore (p \rightarrow q)$ is True.

$\therefore$ If $3n+2$ is an even integer, then n is an even integer. $\square$

2) Prove the proposition $P(0)$, where $P(n)$ is the proposition "If n is a positive integer greater than 1, then $n^2 > n$." What kind of proof did you use?

Solution: $P(0)$: If 0 is a positive integer greater than 1, then $0^2 > 0$.

The proposition that $0 > 1$ is False. $\therefore$ By defn. of conditional, $P(0)$ is True. This is an example of a vacuous proof $\square$

2) Prove the proposition $P(1)$, where $P(n)$ is the proposition "If n is a positive integer, then $n^2 > n$." What kind of proof did you use?

Solution: $P(1)$: If 1 is a positive integer, then $1^2 > 1$.